SEA
Simple Enrichment Analysis

For further information on how to interpret these results please access https://meme-suite.org/meme/doc/sea-output-format.html.
To get a copy of the MEME software please access https://meme-suite.org.

If you use SEA in your research, please cite the following paper:
Timothy L. Bailey and Charles E. Grant, "SEA: Simple Enrichment Analysis of motifs", *BioRxiv*, August 24, 2021. [full text]

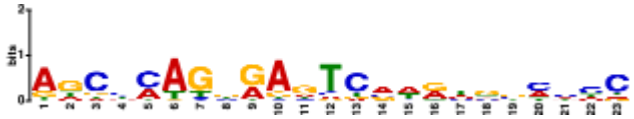
Enriched Motifs | Input Files | Program information | Results in TSV Format | Matching Sequences | Matching Sites

Enriched Motifs

Found 17 motifs with *E*-values ≤ 10.

Logo	Database	ID	I
	H12CORE meme format	ZN362.H12CORE.0.P.C	
	H12CORE meme format	IRF4.H12CORE.0.P.B	
	H12CORE meme format	PO2F3.H12CORE.1.S.C	
	H12CORE meme format	PO3F1.H12CORE.3.M.C	
	H12CORE meme format	PO3F2.H12CORE.2.SM.B	

Logo	Database	ID	A I
	H12CORE meme format	PO3F4.H12CORE.0.SM.B	
	H12CORE meme format	KLF17.H12CORE.1.P.C	
	H12CORE meme format	NR1D1.H12CORE.1.SM.B	
	H12CORE meme format	SPIB.H12CORE.1.S.C	
	H12CORE meme format	ZN132.H12CORE.0.P.C	
	H12CORE meme format	IRF9.H12CORE.0.PSM.A	
	H12CORE meme format	PO3F1.H12CORE.0.P.B	
	H12CORE meme format	THB.H12CORE.1.PSM.A	

Logo	Database	ID
	H12CORE meme format	VENTX.H12CORE.1.S.C
	H12CORE meme format	ZKSC8.H12CORE.1.P.B
	H12CORE meme format	ZN224.H12CORE.0.P.B
	H12CORE meme format	ZNF34.H12CORE.0.P.C

Input Files

Alphabet

Background source: built from the control (negative) sequences

Order: 2 (only order-0 shown)

Name	Bg.				Bg.	Name
Adenine	0.2529	A	~	T	0.2529	Thymine
Cytosine	0.2471	C	~	G	0.2471	Guanine

Sequences

Primary Sequences	Number	Control Sequences	Number
data_s1_H3K4me3.bed.fa	10	primary sequences shuffled conserving 3-mer frequencies	10

Motifs

Database	Source	Motif Count
H12CORE meme format	motif_db/HOCOMOCO/H12CORE_meme_format.meme	1443

Other Settings

Strand Handling	Both the given and reverse complement strands are processed.
Objective Function	Differential Enrichment
Statistical Test	Fisher Exact Test

Sequence Shuffling	Control sequences are primary sequences shuffled preserving 3-mer frequencies.
Hold-out Set	There were not enough input sequences to create a hold-out set so the *p*-values will be less accurate.
Pseudocount	0.1 times the background frequencies
Significance threshold	Motifs must have an enrichment *E*-value of 10 or smaller.
Random Number Seed	0
Trimming of Control Sequences	Trimming of control sequences was allowed.

SEA version
5.5.8 (Release date: Thu May 15 15:01:46 2025 -0700)

Reference
Timothy L. Bailey and Charles E. Grant, "SEA: Simple Enrichment Analysis of motifs", *BioRxiv*, August 24, 2021. [full text]

Command line summary

```
sea --verbosity 1 --oc . --thresh 10.0 --align center --p data_s1_H3K4me3.bed.fa --m
motif_db/HOCOMOCO/H12CORE_meme_format.meme
```